CLINICAL · MOUTH & THROAT

# Oesophagus and dysphagia

Dysphagia is a symptom with a long list of causes, from a stroke to a stricture to a cancer. Most are sorted by asking three questions. Is the problem getting food out of the mouth and pharynx (oropharyngeal), or getting it down the oesophagus (oesophageal)? Is it worse for solids or for liquids? Is it progressive or intermittent? Progressive dysphagia to solids, then liquids, with weight loss in an older smoker is cancer until endoscopy proves otherwise.

**AT A GLANCE**

- Constrictions at 15, 25 and 40 cm from the incisors.
- Achalasia: liquids and solids, bird beak, exclude a cardia cancer.
- Caustics: alkali penetrates deep; stricture from week 3.
- Pain after oesophagoscopy is perforation until proved otherwise.

## 1 CHAPTER ONE Applied anatomy

A 25-centimetre muscular tube.

### 1.1 Course and narrowings

The oesophagus is about 25 cm long. It runs from the cricopharyngeus at C6 to the cardia at T11, deviating slightly to the left in the neck and again in the lower chest. Along the way it has three normal constrictions, measured at endoscopy from the upper incisors:

1. The pharyngo-oesophageal junction at C6, about *15 cm*.
2. The crossing of the aortic arch and left main bronchus at T4, about *25 cm*.
3. The diaphragmatic hiatus at T10, about *40 cm*, with the cardia at about 40 to 42 cm.

These narrowings are where foreign bodies lodge, where caustic burns are worst and where instruments perforate, so their distances are worth knowing by heart.

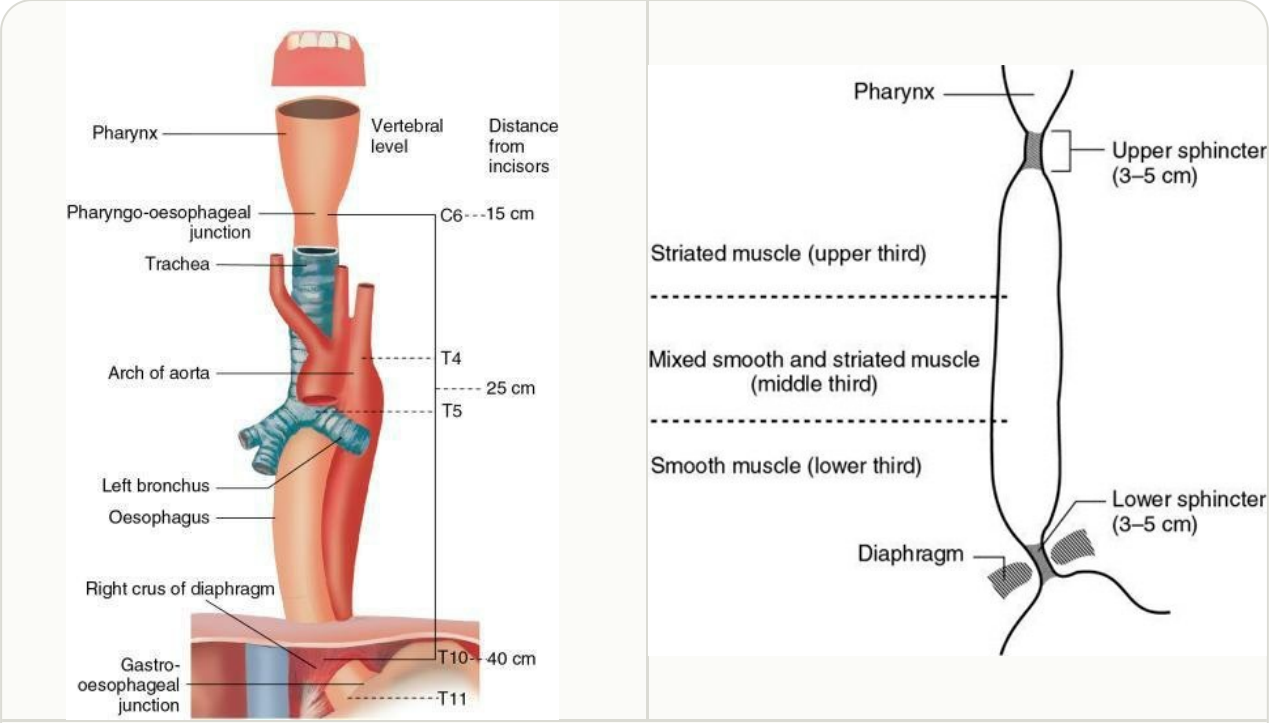



**FIGURE 1** First, the oesophagus and its normal constrictions, with distances from the upper incisors. Second, the oesophagus with its sphincters and the change from striated to smooth muscle.

Dhingra, figs 67.1 and 67.2

## 1.2 Wall and supply

The wall has a mucosa of non-keratinised stratified squamous epithelium, a submucosa carrying *Meissner's* plexus, and inner circular and outer longitudinal muscle with *Auerbach's (myenteric)* plexus between them. Outside there is only adventitia. The absence of a *serosa* is the reason leaks spread freely through the mediastinum and anastomoses are fragile.

The muscle is striated in the upper third, mixed in the middle third and smooth in the lower third, which explains why diseases of skeletal muscle affect the top and scleroderma the bottom. The upper third is supplied by the inferior thyroid artery, the middle by branches of the aorta, and the lower by the left gastric and inferior phrenic arteries. The lower oesophagus is a portosystemic anastomosis and the site of *varices*. Lymph drains to deep cervical, mediastinal and coeliac nodes, and a long submucosal lymphatic network allows *skip lesions* in cancer. The nerve supply is from the vagus and the sympathetic chain.

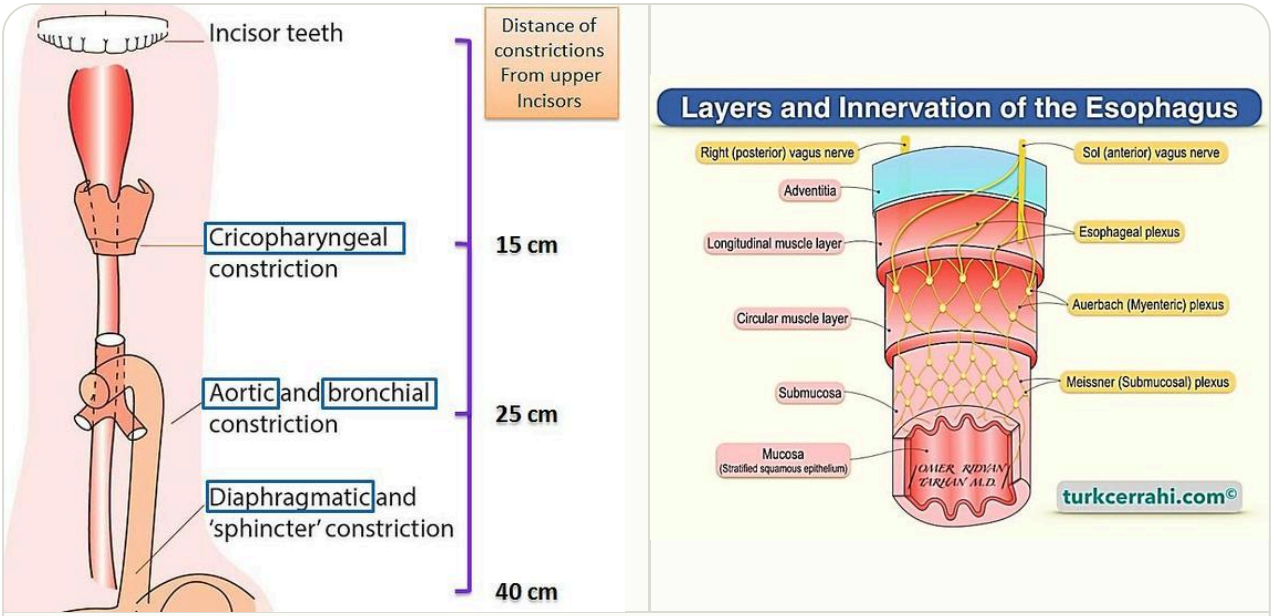



**FIGURE 2** Constrictions of the oesophagus and their distances from the incisors, and the layers and innervation of the oesophagus.

Anatomy atlas · ENT Steps p. 226, ENT Steps p. 227

## 1.3 Sphincters

The **upper oesophageal sphincter** is mainly the *cricopharyngeus*, a zone 3 to 5 cm long that stays tonically closed at rest to stop air entering the oesophagus and refluxate entering the pharynx. On swallowing it relaxes and is pulled open by elevation of the larynx. Above it, between the thyropharyngeus and the cricopharyngeus, lies *Killian's dehiscence*, the site of the pharyngeal pouch; below it, between the cricopharyngeus and the oesophageal muscle, lies *Laimer's triangle*.

The **lower oesophageal sphincter** is a physiological high-pressure zone rather than an anatomical sphincter. It is helped by the "pinch-cock" of the diaphragmatic crura, the acute *angle of His*, the mucosal folds, and the positive intra-abdominal pressure acting on the abdominal segment.

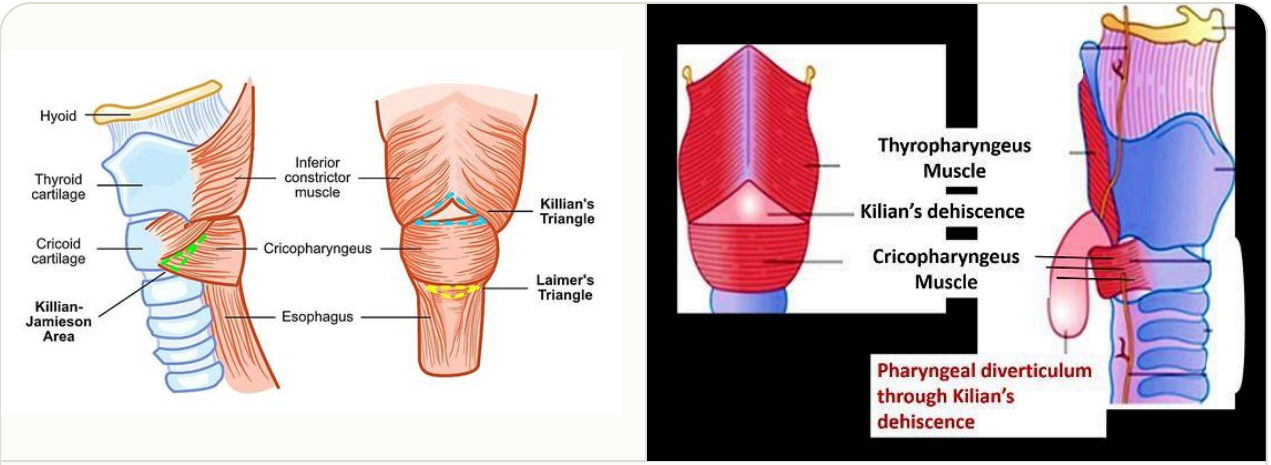



**FIGURE 3** Killian and Laimer triangles, and Killian dehiscence with a pharyngeal pouch.
Anatomy atlas · ENT Steps p. 225, ENT Steps p. 198

# 2 CHAPTER TWO Swallowing

Three phases.

## 2.1 The phases

The **oral phase** is voluntary: food is chewed, mixed with saliva, formed into a bolus and pushed back by the tongue against the palate. It ends when the bolus touches the faucial pillars and triggers the reflex swallow.

The **pharyngeal phase** is a reflex lasting under a second, a sequence of protective closures and propulsion. The soft palate rises against *Passavant's ridge* to close the nasopharynx, and the tongue presses against the palate to close the mouth. Breathing stops and the larynx closes from below upwards, at the true folds, then the false folds, then the arytenoids against the epiglottis. The larynx rises and moves forward under the tongue base, which tips the epiglottis down and *pulls the upper sphincter open*; laryngeal elevation is the single most important factor in opening it. The constrictors then contract in sequence while the cricopharyngeus relaxes.

In the **oesophageal phase**, peristalsis carries the bolus to the stomach in about 3 to 8 seconds, with the lower sphincter relaxing ahead of it.

### 2.2 Peristalsis

**Primary peristalsis** is the wave started by a swallow. **Secondary peristalsis** is set off locally by distension and clears what the first wave left behind. **Tertiary contractions** are non-propulsive and uncoordinated; they appear in the elderly (*presbyoesophagus*) and in spasm.

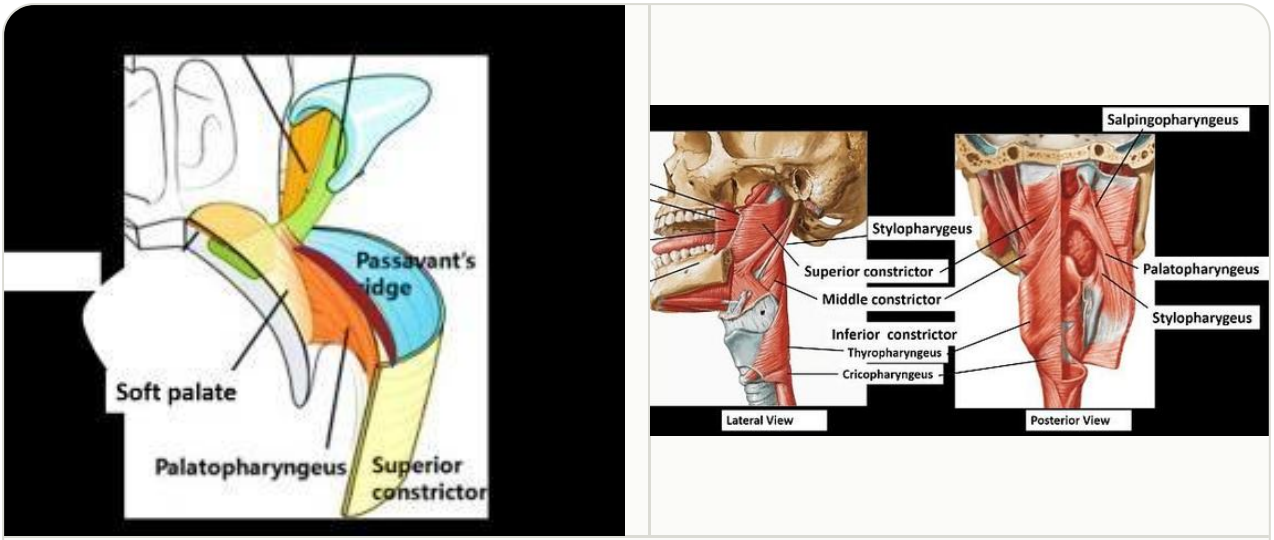



**FIGURE 4** Passavant ridge, and the pharyngeal constrictors.
Anatomy atlas · ENT Steps p. 205, ENT Steps p. 197

## 3 CHAPTER THREE Approach to dysphagia

Where, what and how fast.

### 3.1 Terms

**Dysphagia** is difficulty swallowing. **Odynophagia** is painful swallowing and suggests ulceration, inflammation or infection. **Globus** is a persistent feeling of a lump in the throat, worst between meals, with no true difficulty in swallowing food.

## 3.2 Oropharyngeal versus oesophageal

TWO KINDS OF DYSPHAGIA

| FEATURE | OROPHARYNGEAL (PRE-OESOPHAGEAL) | OESOPHAGEAL |
|---|---|---|
| Complaint | Difficulty starting a swallow; food sticks in the neck | Food sticks behind the sternum (though it may be felt in the neck) |
| Associated | Coughing or choking on swallowing, nasal regurgitation, drooling, dysarthria, a wet voice | Regurgitation, heartburn, chest pain, weight loss |
| Typical causes | Stroke, Parkinson disease, motor neurone disease, myasthenia, myositis, cranial nerve palsies, head and neck cancer and its treatment, pharyngeal pouch, cricopharyngeal dysfunction | Stricture, cancer, achalasia, spasm, eosinophilic oesophagitis, rings and webs, extrinsic compression |
| First test | Examination, flexible endoscopy, videofluoroscopy or FEES | Endoscopy (upper GI), barium swallow, manometry |

## 3.3 Causes by site

It helps to think of the tube from the inside out. **In the lumen** lies a foreign body or food bolus. **In the wall** are inflammation (reflux, infection, caustic, eosinophilic, radiation or pill oesophagitis), strictures, webs and rings, tumours, and motility disorders. **Outside the wall** are the pharyngeal pouch, goitre and thyroid tumours, cervical *osteophytes* in diffuse idiopathic skeletal hyperostosis, mediastinal nodes and tumours, an enlarged left atrium, aortic aneurysm, and vascular rings causing *dysphagia lusoria*.

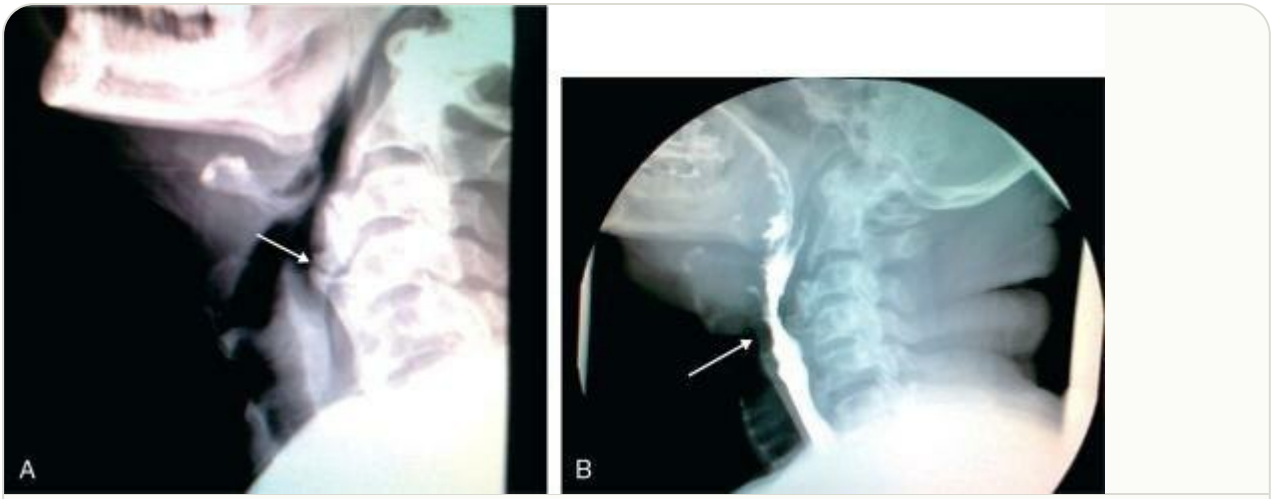



**FIGURE 5** (A and B) Large cervical osteophytes pressing on the oesophagus on barium swallow.
Dhingra, fig. 69.1

## 3.4 History

The **tempo** points to the cause. Sudden dysphagia means a foreign body or bolus, often stuck on an existing stricture or tumour; progression over weeks to months means cancer; intermittent symptoms suggest a ring or spasm; and long-standing dysphagia suggests achalasia or a web.

The **consistency** separates mechanical from motor disease. Solids first and liquids later indicate a mechanical obstruction such as a stricture or cancer, whereas liquids as bad as or worse than solids from the start indicate a motility disorder such as achalasia, or a neuromuscular cause.

Other clues complete the picture. Night-time regurgitation of undigested food with gurgling and halitosis suggests a pharyngeal pouch; heartburn suggests reflux and a peptic stricture; nasal regurgitation suggests palatal weakness; and coughing on swallowing means aspiration. Odynophagia suggests candida or herpes in the immunosuppressed, a pill or a caustic. Weight loss, hoarseness or a neck mass suggests cancer, and recurrent bolus impaction in a young atopic man suggests eosinophilic oesophagitis.

## 3.5 Examination and tests

Examine the mouth and oropharynx, and look at the larynx and hypopharynx with a flexible endoscope for pooling in the pyriform sinuses and for cord movement; then

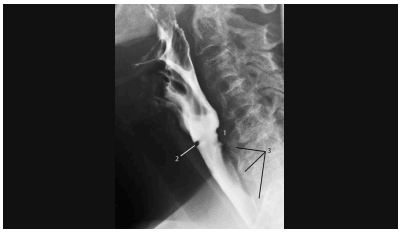
examine the neck, the cranial nerves and the chest. A blood count and iron studies look for iron deficiency and Plummer–Vinson syndrome.

The **barium swallow** shows pouches, strictures, webs, rings, achalasia, spasm and extrinsic compression. **Videofluoroscopy**, the modified barium swallow, studies the oral and pharyngeal phases with different consistencies and demonstrates aspiration, and **flexible endoscopic evaluation of swallowing (FEES)** shows residue, penetration and aspiration at the bedside. A chest film, a lateral soft-tissue neck film and CT are used for masses.

**Endoscopy**, either oesophagogastroduodenoscopy or transnasal oesophagoscopy, sees and biopsies the mucosa, and *every new oesophageal dysphagia in an adult needs it.* For function, **high-resolution manometry** is the test for motility disorders, classified by the *Chicago classification*, and **24-hour pH and impedance monitoring** quantifies reflux.

IMAGING IN THE APP 1 case

Disorders of the oesophageal passage on imaging.



**Esophageal Passage Disorders**

Infection & inflammation · 2 images

# 4 CHAPTER FOUR Inflammation and injury

Reflux, infection, caustics, perforation.

## 4.1 Gastro-oesophageal reflux disease

Reflux disease is a failure of the barrier, through weakness or transient relaxation of the lower sphincter, hiatus hernia, obesity, pregnancy, smoking, alcohol, fatty meals,

scleroderma, and drugs that relax smooth muscle such as calcium-channel blockers, nitrates and anticholinergics. Its typical symptoms are heartburn and regurgitation.

For the ENT surgeon the extra-oesophageal effects matter as much. *Laryngopharyngeal reflux* causes hoarseness, throat clearing, globus and chronic cough, with posterior laryngitis, contact ulcers and granulomas; reflux also contributes to subglottic stenosis, asthma, dental erosion and, in children, otitis media with effusion.

The complications are erosive oesophagitis, peptic stricture and *Barrett's oesophagus*, in which columnar *intestinal metaplasia* replaces the squamous lining of the distal oesophagus. It appears as salmon-pink mucosa, is premalignant for *adenocarcinoma*, and needs endoscopic surveillance.

Treatment starts with lifestyle change: weight loss, raising the head of the bed, no meals in the three hours before bed, stopping smoking, and less alcohol, caffeine, chocolate and fat. Proton pump inhibitors, H2 blockers and alginates follow, and antireflux surgery such as a *Nissen fundoplication* is reserved for refractory disease.

## 4.2 Infective oesophagitis

Infection occurs in the immunosuppressed, after chemotherapy or steroids, in HIV and in diabetes, and presents with odynophagia. **Candida** produces white plaques and is treated with fluconazole; **herpes simplex** produces small punched-out ulcers and is treated with aciclovir; and **cytomegalovirus** produces large shallow ulcers and is treated with ganciclovir.

## 4.3 Eosinophilic oesophagitis

Eosinophilic oesophagitis typically affects young atopic men with asthma, eczema or food allergy, and presents with dysphagia and *recurrent food bolus impaction*. Endoscopy shows rings ("trachealisation"), furrows and white exudates, and biopsy shows *15 or more eosinophils per high-power field*. It is treated with a PPI, swallowed topical steroids such as budesonide or fluticasone, elimination diets and careful dilatation.

## 4.4 Pill and radiation oesophagitis

Tetracyclines and doxycycline, potassium chloride, iron, NSAIDs and bisphosphonates can lodge at the aortic narrowing and ulcerate it, especially when taken at bedtime without water. Radiation causes acute ulceration and, later, stricture and dysmotility.

## 4.5 Caustic (corrosive) injury

Two patterns are seen: a toddler who swallows a household product by accident, usually in small amounts with less severe injury, and an adolescent or adult who swallows a large amount on purpose, with severe injury. **Alkalis** such as drain cleaner and lye cause *liquefactive necrosis* and penetrate deeply into the oesophagus. **Acids** cause *coagulative necrosis*, which limits their penetration, and damage the stomach more, leading to pyloric stenosis or gastric perforation.

The injury evolves in four stages:

1. Acute necrosis in the first days.
2. Sloughing and granulation over two to three weeks, when the wall is weakest and perforation most likely.
3. Fibrosis and *stricture* from about the third week, over months.
4. A lifelong increased risk of *squamous carcinoma*, decades later.

Assessment begins with the airway: stridor, hoarseness and drooling mean laryngeal injury, which may need intubation or tracheostomy. Burns of the lips and mouth are unreliable, because their absence does not exclude oesophageal injury. Look for perforation, mediastinitis or peritonitis, and take chest and abdominal films, with CT if perforation is suspected. Endoscopy is then done gently within 12 to 48 hours, going no further than a severe circumferential burn, and graded by Zargar's system: grade 0 normal, 1 oedema and erythema, 2a superficial ulcers and exudates, 2b deep or circumferential ulcers, 3a focal necrosis and 3b extensive necrosis. Grades 0 to 2a heal without stricture; grade 2b and above often stricture.

Management follows a sequence:

1. Resuscitate and protect the airway.
2. *No induced vomiting, no neutralising agents and no blind nasogastric tube* (most current guidance).

3. Analgesia and acid suppression.
4. Nil by mouth at first; a nasogastric tube placed under vision, or a gastrostomy, for feeding in deep burns.
5. Antibiotics for deep burns or perforation.
6. Steroids are controversial and not used routinely.
7. Emergency oesophagectomy or gastrectomy for full-thickness necrosis or perforation.
8. Follow-up contrast studies for stricture.

Strictures are treated by repeated dilatation, retrograde dilatation through a gastrostomy over a swallowed string when the stricture cannot be passed from above, and oesophageal replacement with colon or stomach when dilatation fails.

WHERE THE BOOKS DIFFER · CORROSIVE BURNS: NEUTRALISATION AND STEROIDS

| Dhingra | Lee |
| --- | --- |
| Neutralisation with a weak acid or alkali within six hours, a nasogastric tube to keep the lumen open, and steroids from 48 to 96 hours for four to six weeks to prevent stricture. | Supportive care, early endoscopic grading, airway protection when the larynx is involved, and management of later strictures, with no routine neutralisation or steroids. |

Current practice avoids neutralisation, because the heat of the reaction adds injury and vomiting re-exposes the oesophagus. Steroids are not routinely given, because trials have not shown that they prevent stricture.

## 4.6 Perforation

Instrumentation is the commonest cause of perforation: rigid oesophagoscopy, dilatation and the removal of foreign bodies, usually just above or at the cricopharyngeus at Killian's dehiscence, or in the distal oesophagus. Foreign bodies, trauma and malignancy are other causes, and *Boerhaave syndrome* is a spontaneous full-thickness rupture after vomiting, usually of the left lower oesophagus. It must be distinguished from the **Mallory–Weiss** tear, a mucosal tear at the gastro-oesophageal junction after retching that presents with haematemesis and is not a perforation.

A **cervical perforation** causes neck pain, dysphagia, fever, tenderness and *surgical emphysema* in the neck. A **thoracic perforation** causes severe chest or *interscapular*

pain, fever, tachycardia and shock, with surgical emphysema, *Hamman's sign* (a mediastinal crunch with each heartbeat), pneumothorax and pleural effusion. *Mackler's triad* of vomiting, chest pain and subcutaneous emphysema is the classic picture of Boerhaave syndrome.

Neck and chest films show air in the prevertebral space, mediastinum or pleura, and a *water-soluble contrast swallow* and CT confirm the leak. Every patient is kept nil by mouth on intravenous fluids, broad-spectrum antibiotics and antifungals, with nutritional support. Small contained cervical perforations are often managed conservatively, with cervical drainage if an abscess forms. Thoracic perforations need early surgical repair, ideally within about 24 hours, with drainage, or endoscopic stents and clips in selected cases; late perforations need drainage, diversion or resection.

**PAIN AFTER AN ENDOSCOPY**

Neck, chest or back pain, fever, tachycardia or crepitus after oesophagoscopy or dilatation means perforation until proved otherwise. Stop oral intake, image with water-soluble contrast and CT, and start antibiotics. Every hour of delay raises mortality.

# 5 CHAPTER FIVE Motility disorders

Manometry makes the diagnosis.

## 5.1 Achalasia

Achalasia results from loss of the inhibitory neurons of the *myenteric (Auerbach's) plexus*, producing *aperistalsis* and failure of the lower sphincter to relax, usually with a high resting pressure. A secondary form occurs in *Chagas disease* from *Trypanosoma cruzi*, and *pseudoachalasia* is produced by a cancer at the cardia, which must always be excluded by endoscopy, especially in an older patient with rapid weight loss.

The patient has dysphagia to *both solids and liquids* from the outset, regurgitates undigested food especially at night with cough and aspiration, has chest pain, and loses weight slowly. The barium swallow shows a dilated oesophagus with a fluid

level tapering smoothly to a "*bird's beak*" at the cardia, and a chest film may show a widened mediastinum and an absent gastric bubble. Manometry is the gold standard, and the Chicago classification divides achalasia into types I to III, which predicts the response to treatment.

The options are *pneumatic dilatation*, laparoscopic *Heller myotomy* with a partial fundoplication, and *peroral endoscopic myotomy (POEM)*. Botulinum toxin injected into the sphincter gives temporary relief in the frail, and calcium-channel blockers and nitrates have only a weak effect. The long-term risk of squamous carcinoma is increased.

## 5.2 Spastic disorders

**Distal (diffuse) oesophageal spasm** produces premature, non-propulsive contractions with normal relaxation of the lower sphincter. It causes intermittent dysphagia and chest pain that mimics angina, and the barium swallow shows a "*corkscrew*" or "*rosary bead*" oesophagus. It is treated with nitrates, calcium-channel blockers, botulinum toxin and sometimes myotomy. The **hypercontractile** ("nutcracker" or "jackhammer") oesophagus has very strong contractions that remain peristaltic, and causes chest pain.

## 5.3 Systemic causes

In **scleroderma** (systemic sclerosis), smooth-muscle atrophy and fibrosis give *aperistalsis of the lower two-thirds* and a *weak lower sphincter*, with severe reflux, strictures and Barrett's oesophagus, as part of the CREST syndrome. Polymyositis and dermatomyositis affect the striated upper oesophagus, and myasthenia gravis, motor neurone disease, diabetes, hypothyroidism and presbyoesophagus also disturb motility.

## 5.4 Cricopharyngeal dysfunction

Here the upper sphincter fails to relax or open, from fibrosis of the muscle, neurological disease such as stroke, Parkinson disease, motor neurone disease or muscular dystrophy, or reflux. Videofluoroscopy shows a posterior *cricopharyngeal bar*, though a bar is also seen in up to a third of asymptomatic elderly people, so it

must be interpreted with the symptoms. It is treated by dilatation, botulinum toxin injection or *cricopharyngeal myotomy*, endoscopic or external; myotomy is avoided in severe reflux.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Dysfunction of the upper oesophageal sphincter, and oesophagoscopy and hypopharyngoscopy.

**Dysfunction of the Upper Esophageal Sphincter**
Upper sphincter dysfunction

**Esophagoscopy and Hypopharyngoscopy**
Oesophagoscopy

CHAPTER SIX

# 6 Structural disorders

Webs, rings, pouches, hernias.

## 6.1 Plummer–Vinson (Paterson–Brown-Kelly) syndrome

This syndrome affects middle-aged women and combines iron-deficiency anaemia with dysphagia from a *postcricoid web*, glossitis, angular stomatitis, koilonychia, atrophic mucosa and sometimes achlorhydria. It carries an increased risk of *postcricoid* and oesophageal carcinoma, about 10% in the older literature. It is treated with iron, with B12 and B6 if deficient, dilatation of the web, and surveillance.

## 6.2 Schatzki ring

A Schatzki ring is a thin mucosal ring at the squamocolumnar junction, often with a small hiatus hernia, in people over 50. It causes intermittent dysphagia to solids and, once the lumen is under 13 mm, the "steakhouse syndrome" of bolus impaction. It is treated by dilatation and a PPI.

## 6.3 Diverticula

The **pharyngeal (Zenker's) pouch** is a pulsion diverticulum through Killian's dehiscence and is covered with the other pharyngeal conditions. A **mid-**

**oesophageal traction diverticulum** is pulled out by inflamed mediastinal nodes, classically tuberculous, and an **epiphrenic diverticulum** is a pulsion diverticulum just above the lower sphincter, usually secondary to a motility disorder.

IMAGING IN THE APP 1 case

A pharyngeal (Zenker) diverticulum on barium swallow.



Zenker Diverticulum (Pharyngeal Pouch)

Infection & inflammation · 9 images

## 6.4 Hiatus hernia

In the **sliding** hernia, about 90% of the total, the gastro-oesophageal junction slides up into the chest, and reflux is common. In the **para-oesophageal (rolling)** hernia the junction stays below the diaphragm while the fundus rolls up beside it; reflux is slight, but the hernia can bleed, obstruct or strangulate, and repair is recommended.

## 6.5 Strictures

Benign strictures may be peptic, caustic, radiation-induced, anastomotic after surgery, pill-induced, the result of a foreign body, eosinophilic or congenital. Every stricture is biopsied to exclude cancer. Treatment is dilatation with bougies, often over a guidewire, or balloons, together with treatment of the cause; stents or surgery are reserved for refractory cases.

## 6.6 Congenital anomalies

**Oesophageal atresia with tracheo-oesophageal fistula** is commonest (about 85%) as *proximal atresia with a distal fistula*. The baby drools, chokes and becomes

cyanosed at the first feed, the abdomen distends, and a nasogastric tube coils in the upper pouch at 9 to 13 cm. It is associated with the *VACTERL* anomalies. The **H-type fistula** (about 4%) presents later with coughing on feeds and recurrent pneumonia. **Vascular rings and an aberrant right subclavian artery** cause dysphagia lusoria, and the latter brings a non-recurrent right laryngeal nerve that is at risk in thyroid surgery.

# 7 CHAPTER SEVEN Tumours

Benign and malignant.

## 7.1 Benign tumours

The **leiomyoma** accounts for about two-thirds of benign tumours. It is a smooth, intramural submucosal mass that causes dysphagia only when large and is best seen with endoscopic ultrasound; mucosal biopsy is avoided, because it scars the plane needed for enucleation. **Fibrovascular polyps** of the upper oesophagus can be regurgitated into the mouth and obstruct the larynx, and papillomas, lipomas, haemangiomas and granular cell tumours also occur.

## 7.2 Oesophageal cancer

**Squamous cell carcinoma** is the commonest type worldwide and in India, and arises mostly in the *middle and upper* oesophagus. Its risk factors are tobacco, smoked and chewed, and alcohol; very hot drinks; dietary nitrosamines and poor diet; *achalasia*; *caustic strictures*; Plummer–Vinson syndrome; tylosis (palmoplantar keratoderma); and a previous head and neck cancer through field change. **Adenocarcinoma** is the commonest type in Western countries, arises in the *distal* oesophagus and cardia, and develops in *Barrett's oesophagus* with reflux and obesity.

The tumour spreads directly into the trachea, bronchi, aorta, pericardium and recurrent laryngeal nerves; through lymphatics to cervical, mediastinal and coeliac nodes, including skip lesions; and through the blood to liver, lung and bone.

The cardinal symptoms are *progressive dysphagia, first to solids and then to liquids*, and *weight loss*. Odynophagia and back pain suggest spread outside the wall, hoarseness means recurrent nerve involvement, and coughing on swallowing means a *tracheo-oesophageal fistula*. A sudden bolus obstruction may be the first sign.

Diagnosis is by endoscopy and biopsy. The barium swallow shows an irregular "*apple-core*" narrowing with shouldering, and staging uses CT of chest and abdomen, *PET-CT* and endoscopic ultrasound for depth and nodes, with bronchoscopy for tumours near the airway.

Early tumours are resected endoscopically. Resectable locally advanced disease is treated with neoadjuvant chemoradiotherapy (the CROSS regimen) or perioperative chemotherapy followed by oesophagectomy; definitive chemoradiotherapy is a standard option for squamous carcinoma, particularly in the upper and cervical oesophagus. The stomach is brought up as a gastric pull-up to reconstruct, while the cervical oesophagus after pharyngo-oesophagectomy is replaced with a free jejunal flap or a tubed radial forearm or anterolateral thigh flap. **Cervical oesophageal cancer** is often an extension of postcricoid or hypopharyngeal cancer and may need a *pharyngolaryngo-oesophagectomy* with gastric pull-up.

For palliation, self-expanding metal *stents*, brachytherapy, laser or argon plasma ablation and a feeding gastrostomy or jejunostomy relieve dysphagia, and a covered stent seals a tracheo-oesophageal fistula. The prognosis is poor: overall five-year survival is about 15 to 20%, because more than half of patients have unresectable or metastatic disease at diagnosis.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Cancer of the cervical oesophagus.

**Cancer of the Cervical Esophagus**
Cervical oesophageal cancer

# 8 CHAPTER EIGHT Oesophagoscopy

Rigid, flexible and transnasal.

## 8.1 Rigid oesophagoscopy

Rigid oesophagoscopy is still the instrument for sharp or impacted foreign bodies, for biopsy of cricopharyngeal and upper oesophageal lesions, for dilatation, for endoscopic pouch surgery and for panendoscopy in head and neck cancer. It is contraindicated by cervical spine disease or injury, trismus, a receding jaw, aortic aneurysm and severe systemic illness.

1. General anaesthesia with the tube to the left of the mouth. Supine with the neck flexed and the head extended (the "sniffing" position), so the mouth, pharynx and oesophagus align. Protect the upper teeth.
2. Pass the scope down the right side of the tongue to the arytenoids, then behind the cricoid in the midline.
3. Open the cricopharyngeus by *gentle, sustained lifting*, never force. This is where perforation happens.
4. Keep the lumen in view at all times. Aortic pulsation marks the 25 cm level. The red, velvety gastric mucosa marks the cardia.
5. Inspect again on withdrawal.

The complications are injury to the teeth, lips and arytenoids, tracheal compression in children, and *perforation*, which is why pain, fever and emphysema after the procedure are taken seriously.

## 8.2 Flexible and transnasal oesophagoscopy

**Flexible oesophagoscopy** is done under topical anaesthesia or sedation in the left lateral position through a mouth guard. It sees the oesophagus, stomach and duodenum; allows biopsy, dilatation, banding, stents and removal of soft foreign bodies; is safe in neck and jaw disease; and has largely replaced the rigid scope for diagnosis. **Transnasal oesophagoscopy** is an unsedated office examination with a thin scope passed through the nose, used for dysphagia, reflux and Barrett's screening, the search for a second primary, and tracheo-oesophageal puncture after laryngectomy.



**FIGURE 6** Flexible oesophagoscopy in the left lateral position, through a mouth guard, viewing on a monitor.

Dhingra, fig. 93.1

### RIGID VERSUS FLEXIBLE OESOPHAGOSCOPY

| FEATURE | RIGID | FLEXIBLE |
|---|---|---|
| Anaesthesia | General | Topical with or without sedation |
| Neck and jaw disease | Difficult or contraindicated | Possible |
| Reach | Oesophagus | Oesophagus, stomach, duodenum |
| Foreign bodies | Best for sharp and impacted objects; can sheath sharp points | Soft, blunt objects; needs an overtube for sharp ones |
| Biopsy of the upper sphincter region | Excellent | Can miss the postcricoid area |
| Setting | Inpatient theatre | Day case or office |

# 9 CHAPTER NINE For the examination

What the examiner wants to hear.

## 9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three constrictions** and their distances; Killian dehiscence.
- **The three phases of swallowing** and the role of laryngeal elevation.
- **Oropharyngeal versus oesophageal dysphagia;** solids versus liquids; the investigation sequence.
- **Achalasia:** bird's beak, manometry, pseudoachalasia; dilatation, Heller myotomy, POEM.
- **Plummer–Vinson syndrome** and postcricoid cancer.
- **Caustic injury:** alkali versus acid, the stages, grading, and the stricture and cancer risk.
- **Perforation:** causes and signs; Boerhaave syndrome.
- **Barrett oesophagus** and adenocarcinoma; risk factors for squamous carcinoma.
- **Rigid oesophagoscopy:** position, steps, complications.

## 9.2 Common pitfalls

COMMON PITFALLS

- Calling progressive dysphagia in an older patient "globus" without endoscopy.
- Forcing a rigid scope through the cricopharyngeus.
- Missing perforation in a patient with pain after endoscopy.
- Treating achalasia without excluding a cancer at the cardia.
- Inducing vomiting or neutralising after a caustic ingestion.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, Oxford Specialist Handbook ch. 10, Oxford Handbook of ENT ch. 13, Dhingra ch. 67, 68, 69, 93 and K. J. Lee ch. 36, 45. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

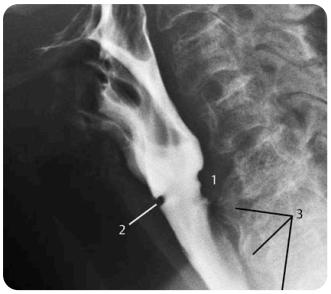
# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

## Esophageal Passage Disorders

- Patients with removable dentures may lack oral sensory input regarding the size of their food. This may also happen with sharp foreign bodies, which may cause perforations of the esophagus.
- Anatomic structures such as the aortic arch or an aberrant right esophageal artery are demonstrated as persistent impressions without stasis of contrast.




Barium swallow, lateral view. Problems with esophageal passage can be a result of a variety of etiologies occurring either singly or in…

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## Zenker Diverticulum (Pharyngeal Pouch)

### IMAGING FINDINGS

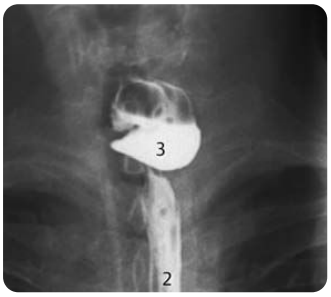
- Outpouching from the posterior hypopharynx above the cricopharyngeus on barium swallow
- Air–fluid level behind the oesophagus on CT
- Residual barium after the swallow

### TOP DIFFERENTIALS

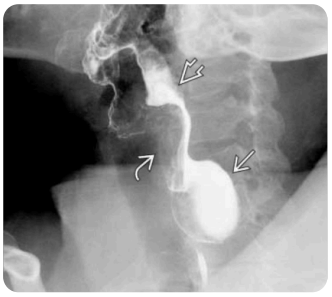
- Lateral pharyngeal or Killian–Jamieson diverticulum
- Paratracheal air cyst

### CHECKLIST

- Size and neck of the pouch
- Aspiration on the swallow




Barium swallow, anteroposterior view. After swallowing contrast, some contrast has remained in the hypopharynx (1) due to physiologic…



Barium esophagram demonstrates a large diverticulum with retained layering contrast from the posterior lateral junction of the hypopharynx…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 258.